

Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

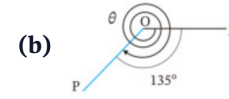
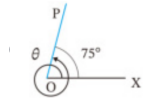
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Solution

1. Add $360^\circ + 75^\circ$ for (a).
2. Use $360^\circ \times (-2) - 135^\circ$ for (b).

Final answer

(a) 435° (b) -855°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q002 Written

Draw the terminal side for 300° and -450° .

Solution

1. Reduce each angle by complete turns while keeping the sign of rotation.
2. 300° ends in quadrant IV; -450° is coterminal with -90° .

Final answer

300° : QIV -450° : negative y – axis

Q003 Written

Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Solution

1. Choose the coterminal angle in the interval $0^\circ \leq \alpha < 360^\circ$.
2. Write the full family by adding $360^\circ \times n$.

Final answer

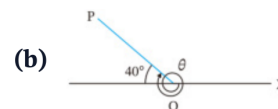
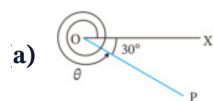
$60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Solution

1. Clockwise rotation is negative, so (a) is -30° .
2. For (b), $180^\circ - 40^\circ = 140^\circ$.

Final answer

(a) -30° (b) 140°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q005 Written

Draw the terminal side for 500° and -120° .

Solution

1. Subtract or add complete turns: $500^\circ - 360^\circ = 140^\circ$; -120° is already in QIII.

Final answer

500° : *QI* -120° : *QIII*

Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Solution

1. Reduce each angle to the principal representative in $[0^\circ, 360^\circ)$.
2. Attach $360^\circ n$.

Final answer

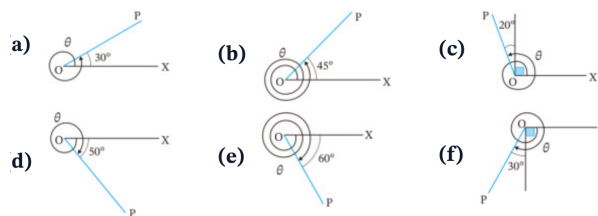
$160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the given figure (a)–(f).



2 Draw the terminal side for the following angles.

Solution

1. Read the marked acute angle and the direction of each complete turn.
2. Use $360^\circ \pm$ marked angle and $90^\circ \pm 20^\circ$ where shown.

Final answer

30° ; 405° ; 110° ; 310° ; 300° ; 240°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .

Solution

1. Reduce each angle modulo 360° , preserving clockwise direction for negative angles.
2. Sketch the ray from the positive x -axis in the resulting quadrant.

Final answer

QII; QII; QIII; QI; QII; QII

Q009 Written

Express 610° , 920° , 840° , -430° , -960° , -20° as $\alpha + 360^\circ \times n$.

Solution

1. Find each representative α in $[0^\circ, 360^\circ)$.
2. The integer coefficient of 360° is absorbed into arbitrary integer n .

Final answer

$250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q010 Written

Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.

Solution

1. Divide by 360° and retain the remainder in the required interval.
2. Write each result as $\alpha + 360^\circ n$.

Final answer

$$120^\circ + 360^\circ n \quad ; \quad 260^\circ + 360^\circ n$$

Q011 MCQ

The principal angle coterminal with 765° is:

Solution

$$1. \quad 765^\circ - 720^\circ = 45^\circ$$

Correct option: A

Final answer

$$45^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Solution**

$$1. -410^\circ + 720^\circ = 310^\circ$$

Correct option: C

Final answer 310°

Q013 MCQ

The terminal side of 540° lies on the:**Solution**

$$1. 540^\circ - 360^\circ = 180^\circ$$

Correct option: B

Final answernegative x -axis

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Solution**1. $315^\circ \in \text{QIV}$

Correct option: D

Final answer

QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Solution**1. $-75^\circ + 360^\circ = 285^\circ$

Correct option: B

Final answer $285^\circ + 360^\circ n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Solution**

1. Negative rotation is clockwise.

Correct option: B

Final answer

clockwise

Q017 MCQ

The terminal side of 1080° is the:**Solution**1. $1080^\circ = 3 \times 360^\circ$

Correct option: A

Final answerpositive x -axis

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Solution**

1. Substitute $\alpha = 0^\circ$ in $\alpha + 360^\circ n$.

Correct option: A**Final answer** $360^\circ n$

Q019 MCQ

The principal angle for -990° is:**Solution**

1. $-990^\circ + 1080^\circ = 90^\circ$

Correct option: C**Final answer** 90°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Solution

1. $180^\circ - 25^\circ = 155^\circ$

Correct option: B

Final answer

155°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Solution

1. Multiply degrees by $\pi/180$, and radians by $180/\pi$.
2. Use $L = r\theta$ and $S = \frac{1}{2}r^2\theta$ with θ in radians.

Final answer

$\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Solution

1. Use the conversion multipliers and the standard table.
2. For a sector, substitute directly in $L = r\theta$ and $S = \frac{1}{2}r^2\theta$.

Final answer

$$\frac{8\pi}{5}, -\frac{20\pi}{9}; 330^\circ, -108^\circ; L = \frac{9\pi}{2}, S = \frac{27\pi}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.

Solution

1. Multiply every degree measure by $\pi/180$.
2. Reduce each fraction to lowest terms.

Final answer

$$\frac{3\pi}{5}, \frac{\pi}{5}, -\frac{4\pi}{3}, -\frac{5\pi}{4}, \frac{7\pi}{3}, \frac{20\pi}{9}, -\frac{19\pi}{6}, -15\pi$$

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ to degrees.

Solution

1. Multiply every radian measure by $180^\circ/\pi$.
2. Keep the sign and simplify.

Final answer

$$210^\circ, 144^\circ, 120^\circ, 240^\circ, -225^\circ, -90^\circ, -990^\circ, -70^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Solution

1. Apply $L = r\theta$.
2. Apply $S = \frac{1}{2}r^2\theta$ to each sector.

Final answer

(a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Solution

1. Use the standard radian-angle table and the sign of rotation.
2. Sketch each terminal ray from the positive x -axis.

Final answer

QI; QI; QII; QII; QIII; QIII; QIV; QIV

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Solution

1. The sector arc becomes the cone base circumference: $2\pi R = 9(4\pi/3)$, so $R = 6$.
2. The slant height is 9, hence $h = \sqrt{9^2 - 6^2} = 3\sqrt{5}$.
3. Use $V = \frac{1}{3}\pi R^2 h$.

Final answer

$$h = 3\sqrt{5} \text{ cm}; \quad V = 36\pi\sqrt{5} \text{ cm}^3$$

Q028 MCQ

Convert 150° to radians:

Solution

1. $150 \times \pi/180 = 5\pi/6$

Correct option: A

Final answer

$$\frac{5\pi}{6}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:

Solution

1. $-5\pi/6 \times 180/\pi = -150^\circ$

Correct option: B

Final answer

-150°

Q030 MCQ

For $r = 5, \theta = \frac{2\pi}{3}$, the arc length is:

Solution

1. $L = r\theta = 5(2\pi/3)$

Correct option: B

Final answer

$\frac{10\pi}{3}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Solution

1. $S = \frac{1}{2}(4^2)(\pi/2) = 4\pi$

Correct option: B

Final answer

4π

Q032 MCQ

The radian measure of a full turn is:

Solution

1. A full circumference subtends 2π radians.

Correct option: B

Final answer

2π

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Solution

1. $7\pi/4 \times 180/\pi = 315^\circ$

Correct option: C

Final answer

315°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Solution

1. $-11\pi/6 + 2\pi = \pi/6$

Correct option: A

Final answer

$\frac{\pi}{6}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3$, $\theta = \pi$, the sector area is:

Solution

1. $S = \frac{1}{2}(3^2)\pi = 9\pi/2$

Correct option: D

Final answer

$$\frac{9\pi}{2}$$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:

Solution

1. $5\pi/3 = 300^\circ$

Correct option: D

Final answer

QIV

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.**Solution**

$$1. 252\pi/180 = 7\pi/5.$$

Final answer

$$\frac{7\pi}{5}$$

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .**Solution**

$$1. L = 8(3\pi/4) = 6\pi.$$

$$2. S = \frac{1}{2}(8^2)(3\pi/4) = 24\pi.$$

Final answer

$$L = 6\pi, \quad S = 24\pi$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution

1. The terminal side is in QIII with reference angle $\pi/3$.
2. Use the signs of x, y on the unit circle.

Final answer

$$-\frac{\sqrt{3}}{2}, -\frac{1}{2}, \sqrt{3}$$

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Solution

1. Locate each angle on the unit circle.
2. Read x, y , then use $\sin \theta = y$, $\cos \theta = x$, $\tan \theta = y/x$.

Final answer

$$-\frac{\sqrt{2}}{2}; \frac{1}{2}; \frac{1}{\sqrt{3}}; -\frac{\sqrt{3}}{2}; -1; \text{undefined}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q041 Written

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write “undefined”.

(a) $\sin \frac{\pi}{3}$

(b) $\sin \frac{7}{6}\pi$

(c) $\sin \pi$

(d) $\cos \frac{5}{4}\pi$

(e) $\cos \frac{5}{3}\pi$

(f) $\cos \frac{7}{6}\pi$

(g) $\tan \frac{2}{3}\pi$

(h) $\tan \frac{5}{4}\pi$

(i) $\tan \pi$

(j) $\sin \left(-\frac{\pi}{6}\right)$

(k) $\sin \left(-\frac{5}{4}\pi\right)$

(l) $\sin \left(-\frac{5}{3}\pi\right)$

(m) $\cos \left(-\frac{\pi}{4}\right)$

(n) $\cos \left(-\frac{5}{6}\pi\right)$

(o) $\cos \left(-\frac{4}{3}\pi\right)$

(p) $\tan \left(-\frac{4}{3}\pi\right)$

(q) $\tan \left(-\frac{11}{6}\pi\right)$

(r) $\tan \left(-\frac{3}{2}\pi\right)$

Solution

1. Reduce each angle to a standard reference angle.
2. Apply quadrant signs and the unit-circle exact values; tangent is undefined when $x = 0$.

Final answer

$\frac{\sqrt{3}}{2}; -\frac{1}{2}; 0; -\frac{\sqrt{2}}{2}; \frac{1}{2}; -\frac{\sqrt{3}}{2}; -\sqrt{3}; 1; 0; -\frac{1}{2}; \frac{\sqrt{2}}{2}; \frac{\sqrt{3}}{2}; \frac{\sqrt{2}}{2}; -\frac{\sqrt{3}}{2}; -\frac{1}{2}; -\sqrt{3}; \frac{1}{\sqrt{3}}; \text{undefined}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q042 Written

The panel height is $h(t) = 3 \sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

Solution

1. Substitute $t = 0$: $h(0) = 3 \sin(-\pi/4) + 1.5$.
2. Use $\sin(-\pi/4) = -\sqrt{2}/2$.

Final answer

$$\frac{3(1 - \sqrt{2})}{2} \text{ m}$$

Q043 MCQ

$\sin \frac{\pi}{6}$ equals:

Solution

1. Use the QI special-angle value.

Correct option: B

Final answer

$$\frac{1}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q044 MCQ

 $\cos \frac{2\pi}{3}$ equals:**Solution**1. QII has negative cosine; reference angle $\pi/3$.

Correct option: B

Final answer

$$-\frac{1}{2}$$

Q045 MCQ

 $\tan \frac{3\pi}{4}$ equals:**Solution**1. QII reference angle $\pi/4$.

Correct option: A

Final answer

$$-1$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q046 MCQ

 $\sin(-\frac{\pi}{2})$ equals:**Solution**1. The terminal point is $(0, -1)$.

Correct option: A

Final answer

-1

Q047 MCQ

 $\tan \frac{\pi}{2}$ is:**Solution**1. $\tan = y/x$ and $x = 0$.

Correct option: D

Final answer

undefined

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q048 MCQ

 $\cos \frac{5\pi}{3}$ equals:**Solution**1. $5\pi/3$ has reference angle $\pi/3$ in QIV.

Correct option: D

Final answer

$$\frac{1}{2}$$

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:**Solution**1. By definition, $\sin \theta = y$.

Correct option: B

Final answer

$$y$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q050 MCQ

 $\tan\left(-\frac{3\pi}{4}\right)$ equals:**Solution**1. Tangent is odd: $\tan(-3\pi/4) = -\tan(3\pi/4) = 1$.

Correct option: C

Final answer

1

Q051 MCQ

 $\sin \frac{11\pi}{6}$ equals:**Solution**1. QIV reference angle $\pi/6$, so sine is negative.

Correct option: A

Final answer $-\frac{1}{2}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q052 Written

Find $\sin \frac{5\pi}{6}, \cos \frac{4\pi}{3}, \tan \frac{7\pi}{4}$.**Solution**

1. Use reference angles $\pi/6, \pi/3, \pi/4$ and quadrant signs.

Final answer

$$\frac{1}{2}, -\frac{1}{2}, -1$$

Q053 Written

Find $\sin(-\frac{5\pi}{6}), \cos(-\frac{7\pi}{6}), \tan(-\frac{\pi}{3})$.**Solution**

1. Reduce negative angles and apply odd/even symmetry.

Final answer

$$-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q054 Written

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution

1. Read x, y directly, then compute y/x .

Final answer

$$\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$$

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Solution

1. $3(-1) - 2(-1) = -1$.

Final answer

$$-1$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q056 Written

State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.**Solution**

1. Tangent is undefined when the unit-circle x -coordinate is zero.

Final answer

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.**Solution**

1. Draw the unit circle and the corresponding horizontal/vertical/oblique line.
2. Use the reference angle and select the quadrants with the required sign.

Final answer

$$(a) \frac{7\pi}{6}, \frac{11\pi}{6}; (b) \frac{\pi}{4}, \frac{7\pi}{4}; (c) \frac{2\pi}{3}, \frac{5\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the given Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Solution

1. Find the reference angle from the special triangle.
2. Use the sign of the function to choose the correct quadrants.

Final answer

$$\frac{\pi}{4}, \frac{3\pi}{4}; \frac{5\pi}{6}, \frac{7\pi}{6}; \frac{\pi}{6}, \frac{7\pi}{6}; 0, \pi; \frac{\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{4}, \frac{7\pi}{4}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q059 Written

Solve the sixteen equations in the given exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Solution

1. Apply the same unit-circle/reference-angle method to each equation.
2. Remember that sine/cosine can have two solutions, while $\sin \theta = \pm 1$ and $\cos \theta = \pm 1$ have one in the interval.

Final answer

$\frac{\pi}{6}, \frac{5\pi}{6}; \frac{\pi}{3}, \frac{2\pi}{3}; \frac{\pi}{2}, \frac{3\pi}{2}; 0; 0, \pi; \frac{\pi}{3}, \frac{4\pi}{3}; \frac{\pi}{2}; \frac{5\pi}{4}, \frac{7\pi}{4}; \frac{3\pi}{4}, \frac{5\pi}{4}; \frac{2\pi}{3}, \frac{4\pi}{3}; \frac{5\pi}{6}, \frac{11\pi}{6}; \frac{\pi}{4}, \frac{5\pi}{4}; \frac{4\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{2}; \pi; \frac{\pi}{6}, \frac{11\pi}{6}$

Q060 Written

Solve $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Solution

1. Let $u = \tan \theta$ and factor $u^2 + (1 - \sqrt{3})u - \sqrt{3} = 0$.
2. The two tangent values are 1 and $-\sqrt{3}$; list their angles in $[0, 360^\circ)$.

Final answer

$60^\circ, 135^\circ, 240^\circ, 315^\circ$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Solution

1. Sine is positive in QI and QII with reference angle $\pi/6$.

Correct option: A

Final answer

$$\frac{\pi}{6}, \frac{5\pi}{6}$$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Solution

1. Cosine is negative in QII and QIII.

Correct option: B

Final answer

$$\frac{2\pi}{3}, \frac{4\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:**Solution**

1. Tangent is positive in QI and QIII.

Correct option: A**Final answer**

$$\frac{\pi}{4}, \frac{5\pi}{4}$$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Solution**

1. The unit-circle y-coordinate is zero on the x-axis.

Correct option: A**Final answer**

$$0, \pi$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:**Solution**1. The point $(1, 0)$ corresponds to $\theta = 0$.**Correct option:** A**Final answer**

0

Q066 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:**Solution**1. Tangent is negative in QII/QIV; reference angle $\pi/3$.**Correct option:** B**Final answer** $\frac{2\pi}{3}, \frac{5\pi}{3}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:**Solution**

1. The lowest unit-circle point has y-coordinate -1 .

Correct option: B**Final answer**

$$\frac{3\pi}{2}$$

Q068 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:**Solution**

1. Cosine is the x-coordinate, zero on the y-axis.

Correct option: B**Final answer**

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:

Solution

1. Tangent is zero when $y = 0$ and $x \neq 0$.

Correct option: A

Final answer

$0, \pi$

Q070 Written

Solve $\cos \theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Cosine is positive in QI/QIV with reference angle $\pi/6$.

Final answer

$\frac{\pi}{6}, \frac{11\pi}{6}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Final answer

$$\frac{4\pi}{3}, \frac{5\pi}{3}$$

Q072 Written

Solve $\tan \theta = \frac{1}{\sqrt{3}}$ for $0 \leq \theta < 2\pi$.

Solution

1. Tangent is positive in QI/QIII with reference angle $\pi/6$.

Final answer

$$\frac{\pi}{6}, \frac{7\pi}{6}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.**Solution**1. Rearrange to $\cos \theta = 1/2$, then choose QI/QIV.**Final answer**

$$\frac{\pi}{3}, \frac{5\pi}{3}$$

Q074 Written

Solve $2 \sin \theta + \sqrt{2} = 0$ for $0 \leq \theta < 2\pi$.**Solution**1. Rearrange to $\sin \theta = -\sqrt{2}/2$, then choose QIII/QIV.**Final answer**

$$\frac{5\pi}{4}, \frac{7\pi}{4}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 **Concept Quiz · MCQ**

The radian measure of 240° is:

Solution

1. $240\pi/180 = 4\pi/3$.

Correct option: B

Final answer

$$\frac{4\pi}{3}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Correct option: B

Final answer

$$\frac{4\pi}{3}, \frac{5\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Solution

1. $S = \frac{1}{2}(6^2)(\pi/3) = 6\pi$.

Correct option: C

Final answer

$$6\pi$$

Quiz MCQ 4 Concept Quiz · MCQ

The general-angle family for -300° is:

Solution

1. Add 360° to obtain the principal angle 60° .

Correct option: A

Final answer

$$60^\circ + 360^\circ n$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

Solve $\cos \theta = -\frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Cosine is negative in QII/QIII with reference angle $\pi/4$.

Final answer

$$\frac{3\pi}{4}, \frac{5\pi}{4}$$

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Solution

1. Multiply by $180/\pi$: -390° . Add 720° for the principal representative 330° .

Final answer

$$-390^\circ; 330^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Solution

1. $L = 10(2\pi/5) = 4\pi$.
2. $S = \frac{1}{2}(10^2)(2\pi/5) = 20\pi$.

Final answer

$$L = 4\pi, \quad S = 20\pi$$

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Solution

1. $3(-1/2) - 2(-1/2) = -1/2$.

Final answer

$$-\frac{1}{2}$$

Answer key

Use the question number shown on each page.

Q001 (a) 435° (b) -855°

Q002 300° : QIV

-450° : *negative y - axis*

Q003 $60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Q004 (a) -30° (b) 140°

Q005 500° : QI -120° : QIII

Q006 $160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Q007 30° ; 405° ; 110° ; 310° ; 300° ; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Q010 $120^\circ + 360^\circ n$; $260^\circ + 360^\circ n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Q022 $\frac{8\pi}{5}$, $-\frac{20\pi}{9}$; 330° , -108° ; $L = \frac{9\pi}{2}$, $S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}$, $\frac{\pi}{5}$, $-\frac{4\pi}{3}$, $-\frac{5\pi}{4}$, $\frac{7\pi}{3}$, $\frac{20\pi}{9}$, $-\frac{19\pi}{6}$, -15π

Q024 210° , 144° , 120° , 240° , -225° , -90° , -990° , -70°

Q025 (a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}$ cm; $V = 36\pi\sqrt{5}$ cm³

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi$, $S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $\sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $\frac{1}{\sqrt{3}}$; $-\frac{\sqrt{3}}{2}$; -1 ; *undefined*

Q041 $\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; 0 ; $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $-\frac{\sqrt{3}}{2}$; $-\sqrt{3}$; 1 ; 0 ; $-\frac{1}{2}$; $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q042 $\frac{3(1-\sqrt{2})}{2}$ m

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}$, $-\frac{1}{2}$, -1

Q053 $-\frac{1}{2}$, $-\frac{\sqrt{3}}{2}$, $-\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $-\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}$, $\frac{3\pi}{2}$

Q057 (a) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$; (b) $\frac{\pi}{4}$, $\frac{7\pi}{4}$; (c) $\frac{2\pi}{3}$, $\frac{5\pi}{3}$

Q058 $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{7\pi}{6}$; 0 , π ; $\frac{\pi}{3}$, $\frac{5\pi}{3}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$

Q059 $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{3}$, $\frac{2\pi}{3}$; $\frac{3\pi}{2}$, 0 ; 0 , π ; $\frac{\pi}{3}$, $\frac{4\pi}{3}$; $\frac{\pi}{2}$, $\frac{5\pi}{2}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$; $\frac{2\pi}{3}$, $\frac{4\pi}{3}$; $\frac{5\pi}{6}$, $\frac{11\pi}{6}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{3\pi}{2}$, $\frac{3\pi}{2}$; $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q060 60° , 135° , 240° , 315°

Q061 $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $\frac{\sqrt{2}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}$, $\frac{5\pi}{3}$

Q072 $\frac{\pi}{6}$, $\frac{7\pi}{6}$

Q073 $\frac{\pi}{3}$, $\frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}$, $\frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}$, $\frac{5\pi}{4}$

Quiz WRITTEN 6 -390° ; 330°

Quiz WRITTEN 7 $L = 4\pi$, $S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Solution

1. Use $\sin^2 \theta + \cos^2 \theta = 1$, then choose the sign from the quadrant.
2. Use $\tan \theta = \sin \theta / \cos \theta$ (or $1 + \tan^2 \theta = 1 / \cos^2 \theta$).

Final answer

$$(a) \cos \theta = -\frac{2\sqrt{6}}{5}, \tan \theta = \frac{\sqrt{6}}{12}; \quad (b) \cos \theta = \frac{\sqrt{7}}{7}, \sin \theta = -\frac{\sqrt{42}}{7}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Solution

1. Find the missing squared value from a Pythagorean identity.
2. Select signs using the stated quadrant, then form the remaining ratio.

Final answer

$$(a) \sin \theta = -\frac{\sqrt{6}}{3}, \tan \theta = \sqrt{2}; (b) \cos \theta = \frac{\sqrt{5}}{5}, \sin \theta = -\frac{2\sqrt{5}}{5}; (c) \cos \theta = -\frac{4}{5}, \tan \theta = -\frac{3}{4}; (d) \sin \theta = -\frac{\sqrt{21}}{7}, \cos \theta = -\frac{2\sqrt{7}}{7}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q003 Written

For each of the twelve numbered cases in the given Exercise, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Solution

1. Square the known ratio and use $\sin^2 \theta + \cos^2 \theta = 1$ or $1 + \tan^2 \theta = 1/\cos^2 \theta$.
2. Apply the quadrant sign before simplifying the final ratio.

Final answer

$$(a) \left(-\frac{\sqrt{7}}{4}, \frac{3\sqrt{7}}{7}\right) (b) \left(-\frac{\sqrt{5}}{5}, -\frac{1}{2}\right) (c) \left(\frac{\sqrt{2}}{4}, -\frac{\sqrt{14}}{4}\right) (d) \left(-\frac{\sqrt{5}}{3}, -\frac{2\sqrt{5}}{5}\right) (e) \left(-\frac{\sqrt{15}}{4}, -\sqrt{15}\right) (f) \left(\frac{1}{3}, -\frac{2\sqrt{2}}{3}\right) (g) \left(-\frac{3\sqrt{10}}{10}, \frac{\sqrt{10}}{10}\right) (h) \left(-\frac{\sqrt{5}}{5}, -2\right) (i) \left(-\frac{12}{13}, -\frac{12}{5}\right) (j) \left(\frac{12}{13}, -\frac{5}{12}\right) (k) \left(-\frac{3}{5}, \frac{3}{4}\right) (l) \left(\frac{2\sqrt{5}}{5}, -\frac{\sqrt{5}}{5}\right)$$

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Solution

1. Use $\tan \theta + \cot \theta = (\sin^2 \theta + \cos^2 \theta)/(\sin \theta \cos \theta)$ to obtain $\sin \theta \cos \theta = 1/12$.
2. The interval gives QIII or QIV; take the pair consistent with the stated signs.

Final answer

$$\sin \theta = -\frac{1}{2}, \quad \cos \theta = -\frac{\sqrt{3}}{2} \quad \text{or the swapped pair}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q005 Written

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Solution

1. Use $\cos^2 \theta = 1 - 25/169$ and QII signs.

Final answer

$$-\frac{12}{13}, -\frac{5}{12}$$

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Solution

1. Use the Pythagorean identity and QIII signs.

Final answer

$$-\frac{4}{5}, \frac{4}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q007 Written

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Solution

1. Use a 3-4-5 triangle with positive signs.

Final answer

$$\frac{3}{5}, \frac{4}{5}$$

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$.

Solution

1. Use a 5-12-13 triangle and QIV signs.

Final answer

$$-\frac{5}{13}, \frac{12}{13}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q009 Written

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ **in QIII, find** $\cos \theta$ **and** $\tan \theta$.

Solution

1. The reference angle is $\pi/4$; both sine and cosine are negative in QIII.

Final answer

$$-\frac{\sqrt{2}}{2}, 1$$

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ **in QIII, $\cos \theta$ is:**

Solution

1. QIII makes cosine negative.

Correct option: B

Final answer

$$-\frac{4}{5}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:**Solution**

1. Use a 1-2- $\sqrt{5}$ triangle.

Correct option: A

Final answer

$$\frac{2\sqrt{5}}{5}$$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:**Solution**

1. The reference angle is $\pi/4$ and tangent is negative in QIV.

Correct option: B

Final answer

$$-1$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Solution

1. Reference angle $\pi/3$, sine positive in QII.

Correct option: C

Final answer

$$\frac{\sqrt{3}}{2}$$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Solution

1. The triangle is 7-24-25.

Correct option: A

Final answer

$$\frac{7}{24}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, **with** θ **in QII**, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Solution

1. Square the sum: $1 + 2 \sin \theta \cos \theta = 1/4$.
2. Use $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$, then choose the QII sign for (c).

Final answer

$$(a) -\frac{3}{8} \quad (b) \frac{11}{16} \quad (c) \frac{\sqrt{7}}{2}$$

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in QII**, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Solution

1. Square the given sum to find the product.
2. Use the sum-of-cubes identity and the quadrant sign.

Final answer

$$(a) -\frac{4}{9} \quad (b) \frac{37}{27} \quad (c) \frac{\sqrt{17}}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Solution

1. Square the given sum or difference to obtain the product.
2. Use the sum/difference of cubes and select the sign from the quadrant.

Final answer

$$(1) \sin \theta \cos \theta = \frac{1}{8}, \sin^3 \theta - \cos^3 \theta = \frac{7\sqrt{3}}{16}, \sin \theta + \cos \theta = \frac{\sqrt{5}}{2}; (2) -\frac{1}{8}, \frac{9\sqrt{3}}{16}, -\frac{\sqrt{5}}{2}; (3) \frac{3}{8}, \frac{11}{16}, -\frac{\sqrt{7}}{2}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ **and** θ **in QIII**, **evaluate the exact value of** $\sin^6 \theta - \cos^6 \theta$.

Solution

1. Find $\sin \theta \cos \theta = 3/8$, then $\sin^3 \theta - \cos^3 \theta = 11/16$ and $\sin^3 \theta + \cos^3 \theta = -5\sqrt{7}/16$.
2. Multiply the last two expressions.

Final answer

$$-\frac{55\sqrt{7}}{256}$$

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ **in QII**, **find** $\sin \theta \cos \theta$ **and** $\sin \theta - \cos \theta$.

Solution

1. Square the sum and use QII signs.

Final answer

$$-\frac{5}{18}, \frac{\sqrt{14}}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ **in QI, find** $\sin \theta \cos \theta$ **and** $\sin^3 \theta - \cos^3 \theta$.

Solution

1. Use $(s - c)^2 = 1 - 2sc$, then the difference-of-cubes identity.

Final answer

$$-\frac{1}{8}, \frac{7\sqrt{5}}{16}$$

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Solution

1. Find $sc = -3/8$, then apply the sum-of-cubes identity.

Final answer

$$-\frac{11}{16}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Solution

1. Find $sc = 4/9$, so the square of the sum is $17/9$; use QIV sign.

Final answer

$$\frac{\sqrt{11}}{3}$$

Q023 Written

Given $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Solution

1. The product is zero and the difference-of-cubes factor is $1 + sc$.

Final answer

$$\frac{3}{2}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$$-\frac{3}{8}$$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Solution

1. Use $(s + c)^3 - 3sc(s + c)$.

Correct option: B

Final answer

$$\frac{11}{16}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Solution

1. The sum is negative in QIII.

Correct option: B

Final answer

$$-\frac{\sqrt{7}}{2}$$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Solution

1. Use $(s - c)^2 = 1 - 2sc$.

Correct option: A

Final answer

$$\frac{1}{8}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Solution

1. Use the two cubic factors and multiply.

Correct option: A

Final answer

$$-\frac{55\sqrt{7}}{256}$$

Q029 Written

Prove $\frac{\sin \theta}{1 + \cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}$.

Solution

1. Rewrite $1/\tan \theta = \cos \theta/\sin \theta$.
2. Use the common denominator $(1 + \cos \theta)\sin \theta$, then $\sin^2 \theta + \cos^2 \theta = 1$.

Final answer

$$\text{LHS} = \text{RHS}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; **(b)** $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Solution

1. For (a), multiply numerator and denominator by $\cos \theta$.
2. For (b), multiply the first fraction by $(1 + \sin \theta)/(1 + \sin \theta)$, then simplify.

Final answer

Both identities verified

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the given Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Solution

1. Replace tangent by $\sin \theta / \cos \theta$ where useful.
2. Combine fractions and use $\sin^2 \theta + \cos^2 \theta = 1$.

Final answer

LHS = RHS for (a)+(f)

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Solution

1. Set $A = 1 + \sin(x)$ and $(B = \cos(x))$, combine the two fractions, and simplify the numerator to $(2(1 + \sin(x)))$.
2. Use $(1 - \sin^2 x = \cos^2 x)$ in the denominator to obtain $(2 / \sin(x))$.

Final answer

$$\text{LHS} = 2 \csc x$$

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Solution

1. Cross-multiply and use $1 - \sin^2 \theta = \cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q034 **Written**

Prove $\frac{1}{1+\cos \theta} + \frac{1}{1-\cos \theta} = 2 \sec^2 \theta.$

Solution

1. Combine over $1 - \cos^2 \theta = \sin^2 \theta$, then rewrite.

Final answer

Identity proved

Q035 **Written**

Prove $\frac{\tan \theta}{1+\tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}.$

Solution

1. Multiply numerator and denominator on the left by $\cos \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}.$

Solution

1. Multiply top and bottom by $1 + \cos \theta$, then use the Pythagorean identity.

Final answer

Identity proved

Q037 Written

Prove $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}.$

Solution

1. Rationalise the left denominator and cancel $\cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:

Solution

1. Keep one side fixed and transform the other.

Correct option: B

Final answer

Simplify one side

Q039 MCQ

The identity used to replace $1 - \sin^2 \theta$ is:

Solution

1. From $\sin^2 \theta + \cos^2 \theta = 1$.

Correct option: A

Final answer

$\cos^2 \theta$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q040 MCQ

$\tan \theta$ should be rewritten as:

Solution

1. Use the quotient identity.

Correct option: B

Final answer $\sin \theta / \cos \theta$

Q041 MCQ

To add two trig fractions, first find a:

Solution

1. Fraction addition requires a common denominator.

Correct option: B

Final answer

common denominator

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Solution**

1. The two expressions are shown equal.

Correct option: B**Final answer**

LHS=RHS

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:**Solution**

1. Use the 5-12-13 triangle and the QII sign.

Correct option: A**Final answer** $\frac{5}{13}$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Solution

1. Cosine is negative in QII.

Correct option: A

Final answer

$$-\frac{2\sqrt{5}}{5}$$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Solution

1. The 8-15-17 triangle has negative sine in QIV.

Correct option: B

Final answer

$$-\frac{8}{15}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Solution

1. Reference angle $\pi/3$, with both ratios negative in QIII.

Correct option: D

Final answer

$$-\frac{\sqrt{3}}{2}$$

Q047 MCQ

If $s + c = 1$, then sc equals:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$$0$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Solution

1. Use $(s - c)^2 = 1 - 2sc$.

Correct option: A

Final answer

$$-\frac{1}{2}$$

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Solution

1. Use $(s + c)^3 - 3sc(s + c)$.

Correct option: A

Final answer

$$\frac{19}{27}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Solution

1. Use $(s - c)(1 + sc)$ with $sc = 3/8$.

Correct option: A

Final answer

$$\frac{11}{16}$$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Solution

1. Take the reciprocal of tangent.

Correct option: B

Final answer

$$\frac{\cos \theta}{\sin \theta}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Solution**

1. Rearrange the Pythagorean identity.

Correct option: A**Final answer**

$$\sin^2 \theta$$

Q053 MCQ

When proving an identity, which side should normally be transformed?**Solution**

1. Keep one side fixed and simplify the other to it.

Correct option: B**Final answer**

One side only

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:

Solution

1. Replace $1 - \sin^2\theta$ or $1 - \cos^2\theta$ using the identity.

Correct option: B

Final answer

The Pythagorean identity

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin \theta + \cos \theta = \frac{1}{2}$, then $\sin \theta \cos \theta$ is:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$-\frac{3}{8}$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 [Concept Quiz · MCQ](#)

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Solution

1. Use a 3-4-5 triangle.

Correct option: B

Final answer

$$\frac{4}{5}$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Which identity is always valid where defined?

Solution

1. Use the fundamental quotient identity.

Correct option: B

Final answer

$$1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 Concept Quiz · MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Solution

1. Square the difference and use the QIII sign.

Correct option: B

Final answer

$$-\frac{\sqrt{7}}{2}$$

Quiz WRITTEN 5 Concept Quiz · Written

Given $\cos \theta = -\frac{5}{13}$ in QIII, find $\sin \theta$ and $\tan \theta$.

Solution

1. Use the 5-12-13 triangle and QIII signs.

Final answer

$$-\frac{12}{13}, \frac{12}{5}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written

Given $s + c = \frac{2}{3}$, **find** sc and $s^3 + c^3$.

Solution

1. Square the sum to find sc , then apply the sum-of-cubes identity.

Final answer

$$-\frac{5}{18}, \frac{19}{27}$$

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}$.

Solution

1. Cross-multiply and use $1 - \sin^2 \theta = \cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Solution

1. Use a 1-2- $\sqrt{5}$ triangle and QIV signs.
2. Substitute into both sides of the identity.

Final answer

$$\sin \theta = -\frac{2\sqrt{5}}{5}, \cos \theta = \frac{\sqrt{5}}{5}; 6 = 6$$

Use the question number shown on each page.

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Solution

1. Reduce the angle to its reference angle and keep the quadrant sign.
2. Use the quarter-turn identities: $\sin(\theta + \frac{\pi}{2}) = \cos \theta$, $\sin(\theta + \pi) = -\sin \theta$, and $\sin(\theta + \frac{3\pi}{2}) = -\cos \theta$.

Final answer

(1) $-\sin \frac{\pi}{5}$ (2) 0

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Solution

1. Use evenness of cosine and the reference-angle rules.
2. Replace shifted functions by $\sin \theta$ or $\cos \theta$, then simplify.

Final answer

(1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three given demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Solution

1. Reduce each angle to an acute reference angle and apply the sign in its quadrant.
2. For (3), use the quarter-turn and half-turn identities before collecting terms.

Final answer

$$(1) -\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8} \quad (2) -\frac{1}{\sqrt{3}}, -1 \quad (3a) 0, (3b) -1$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q004 Written

Simplify $\sin(\frac{3\pi}{2} - \theta) \sec(\pi + \theta) - \tan(\theta - \frac{\pi}{2}) \cot(\frac{5\pi}{2} + \theta)$.**Solution**

1. Use $\sin(\frac{3\pi}{2} - \theta) = -\cos \theta$, $\sec(\pi + \theta) = -\sec \theta$, $\tan(\theta - \frac{\pi}{2}) = -\cot \theta$, and $\cot(\frac{5\pi}{2} + \theta) = -\tan \theta$.
2. Both products equal 1, so their difference is 0.

Final answer

0

Q005 Written

Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, **and** $y = \tan \theta$, **and state each period and range.****Solution**

1. Mark one standard cycle, then repeat it periodically.
2. For tangent, draw vertical asymptotes at $\theta = \frac{\pi}{2} + k\pi$.

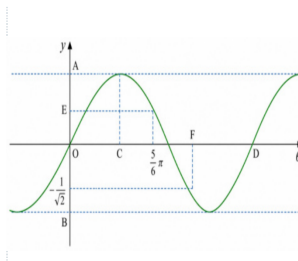
Final answer $\sin, \cos : T = 2\pi, -1 \leq y \leq 1; \quad \tan : T = \pi, y \in \mathbb{R}$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q006 Written

For the given graph of $y = \sin \theta$, determine the labelled scales A to F .

**Solution**

1. Read the extrema and the x-axis scale from the standard sine cycle.
2. Use $\sin \frac{5\pi}{6} = \frac{1}{2}$ and the descending branch to locate F.

Final answer

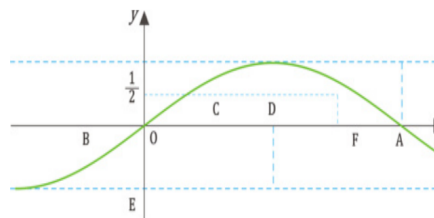
$$A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q007 Written

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the given graph of $y = \cos \theta$, determine the labelled scales A to F .



Solution

1. Use the standard parent graph and mark quarter-period points.
2. The vertical values are the cosine range $[-1, 1]$; the horizontal values follow the 2π cycle.

Final answer

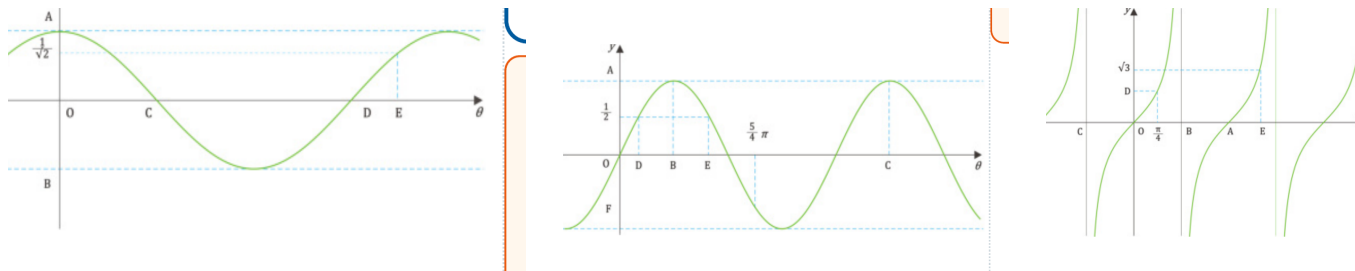
(1) $T_{\sin} = 2\pi$, $T_{\cos} = 2\pi$, $T_{\tan} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q008 Written

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.



Solution

1. Draw axes and the standard cycle first.
2. Read each label from an intercept, extremum, asymptote or quarter-period tick; do not change the graph scale.

Final answer

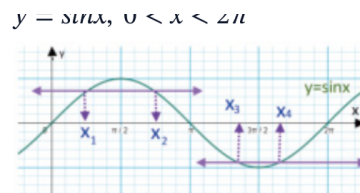
Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q009 Written

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Solution

1. Identify the y-level associated with each marked point and use the sine cycle on $(0, 2\pi)$.
2. Add the four x-coordinates only after they have been read from the same horizontal scale.

Final answer

Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Solution

1. The coefficient 2 outside cosine doubles the y-values but not the period.
2. For $\tan(\theta/2)$, multiply the parent tangent period by the reciprocal of $1/2$.

Final answer

$$y = 2 \cos \theta : T = 2\pi, -2 \leq y \leq 2; \quad y = \tan \frac{\theta}{2} : T = 2\pi$$

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Solution

1. Use the vertical factor for amplitude/range and the reciprocal of the coefficient of θ for horizontal scale.
2. For tangent, the parent period is π ; for sine/cosine it is 2π .

Final answer

$$T = 2\pi, \frac{\pi}{2}, 4\pi \text{ respectively; ranges } [-2, 2], \mathbb{R}, [-1, 1]$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine given graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Solution

1. Read the vertical coefficient as the amplitude/range factor.
2. Divide the parent period by the absolute coefficient of θ .

Final answer

(a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q013 Written

Draw $y = -2|\sin(3x)|$ and find its period.**Solution**

1. The absolute value makes the sine pattern repeat after π in its input, so the x-period is $\pi/3$.
2. The negative outside factor reflects the nonnegative $|\sin(3x)|$ graph below the x-axis.

Final answer

$$T = \frac{\pi}{3}, \quad 0 \geq y \geq -2$$

Q014 Written

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.**Solution**

1. Start the standard sine cycle at $\theta = \pi/6$.
2. A horizontal shift does not change the parent period.

Final answer

$$\text{shift right by } \frac{\pi}{6}, \quad T = 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q015 **Written**

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Solution

1. Read $\theta - p$ as a shift right by p and $\theta + p$ as a shift left by p .
2. Keep the parent period unchanged.

Final answer

Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Solution

1. Translate the parent graph horizontally by the signed amount inside the bracket.
2. Use the same vertical range and asymptote spacing as the parent function.

Final answer

All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos |x + \frac{\pi}{3}|$ and find its period.

Solution

1. Use $\cos |u| = \cos(u)$, then read the horizontal shift and vertical scale.
2. The period of cosine remains 2π .

Final answer

$$y = 3 \cos(x + \frac{\pi}{3}) \text{ since cosine is even; } T = 2\pi$$

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Solution

1. Factor the coefficient of θ from the bracket before reading the shift.
2. Apply shift, horizontal scale, then vertical scale.

Final answer

$$(1) y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2; \quad (2) y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Solution

1. Identify the coefficient b , factor the angle, then read the shift q/b .
2. Use the parent period divided by $|b|$ and the outside amplitude $|a|$.

Final answer

$$(a) T = 4\pi, -3 \leq y \leq 3; \quad (b) T = \frac{\pi}{2}; \quad (c) T = \pi, -2 \leq y \leq 2$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine given graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Solution

1. Rewrite every angle as $b(\theta - q/b)$.
2. Apply the transformations from the inside out and record period, range and shift.

Final answer

Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q021 **Written****Draw $y = 3 \sin |2x - \pi|$ and find its period.****Solution**

1. The absolute value makes the sine pattern repeat after π in its input; with coefficient 2 the x-period is $\pi/2$.
2. The graph is nonnegative with maximum 3.

Final answer

$$y = 3|\sin(2x - \pi)|, \quad T = \frac{\pi}{2}, \quad 0 \leq y \leq 3$$

Q022 **MCQ** **$\sin(\theta + \pi)$ equals:****Solution**

1. A half-turn changes the sign of sine.

Correct option: B**Final answer**

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:**Solution**

1. A half-turn changes the sign of cosine.

Correct option: B

Final answer

$$-\cos \theta$$

Q024 MCQ

 $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:**Solution**

1. Use the complementary-angle identity.

Correct option: C

Final answer

$$\cos \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:**Solution**

1. Cosine is even.

Correct option: A

Final answer $\cos \theta$

Q026 MCQ

 $\sin(-\theta)$ equals:**Solution**

1. Sine is odd.

Correct option: B

Final answer $-\sin \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:**Solution**

1. Use the supplementary-angle identity.

Correct option: B

Final answer

$$-\cos \theta$$

Q028 MCQ

 $\sin\left(\frac{3\pi}{2} + \theta\right)$ equals:**Solution**

1. Shift by three quarter-turns.

Correct option: D

Final answer

$$-\cos \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:**Solution**

1. Use the quarter-turn tangent identity.

Correct option: D

Final answer $-\cot \theta$

Q030 MCQ

The period of $y = \sin \theta$ is:**Solution**1. One sine cycle spans 2π .

Correct option: C

Final answer 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Solution**

1. Cosine is bounded by -1 and 1.

Correct option: B

Final answer

$$-1 \leq y \leq 1$$

Q032 MCQ

The period of $y = \tan \theta$ is:**Solution**1. Tangent repeats every π .

Correct option: B

Final answer

$$\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Solution**

1. Cosine is zero at odd multiples of $\pi/2$.

Correct option: B**Final answer**

$$\theta = \frac{\pi}{2} + k\pi$$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Solution**

1. Read the top of the standard sine graph.

Correct option: C**Final answer**

1

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Solution**

1. Sine first reaches 1 at $\pi/2$.

Correct option: B**Final answer** $\pi/2$

Q036 MCQ

At $\theta = 0$, $\cos \theta$ equals:**Solution**

1. The cosine graph starts at its maximum.

Correct option: C**Final answer**

1

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Solution**

1. Tangent passes through the origin.

Correct option: B**Final answer**

0

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Solution**

1. Multiply the sine range by 3.

Correct option: B**Final answer** $-3 \leq y \leq 3$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Solution**1. Use $2\pi/|b|$.**Correct option:** B**Final answer** $\pi/2$

Q040 MCQ

The period of $y = \cos(\theta/3)$ is:**Solution**1. The coefficient is $1/3$, so multiply the parent period by 3.**Correct option:** D**Final answer** 6π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Solution**

1. Use $\pi/|b|$.

Correct option: A**Final answer**

$$\pi/5$$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:**Solution**

1. The outside factor halves the cosine values.

Correct option: C**Final answer**

$$-\frac{1}{2} \leq y \leq \frac{1}{2}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Solution**

1. The negative sign reflects; 2 stretches.

Correct option: B**Final answer**

reflected and stretched vertically

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Solution**1. Multiply the tangent period by the reciprocal of $1/2$.**Correct option:** C**Final answer** 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Solution**

1. Use $2\pi/3$.

Correct option: B**Final answer** $2\pi/3$

Q046 MCQ

The graph of $y = \sin(\theta - \pi/4)$ is shifted:**Solution**

1. $\theta - p$ means shift right by p .

Correct option: B**Final answer**right by $\pi/4$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Solution**1. $\theta + p$ means shift left by p .

Correct option: B

Final answerleft by $\pi/3$

Q048 MCQ

A horizontal shift changes the sine period from 2π to:**Solution**

1. Translation does not change period.

Correct option: B

Final answer 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Solution**

1. Translation does not change period.

Correct option: B**Final answer** π

Q050 MCQ

The graph of $y = \sin(\theta + \pi/2)$ equals:**Solution**

1. Use the quarter-turn identity.

Correct option: C**Final answer** $\cos \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Solution**1. A π shift reverses cosine.

Correct option: B

Final answer

$$-\cos \theta$$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:**Solution**1. Set $\theta - \pi/2 = 0$.

Correct option: B

Final answer

$$\theta = \pi/2$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Solution**

1. Evenness removes the absolute value inside cosine.

Correct option: A**Final answer** $\cos x$

Q054 MCQ

In $y = a \sin(b\theta - q)$, the horizontal shift is:**Solution**

1. Factor b from the angle first.

Correct option: B**Final answer** q/b

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Solution**

1. The coefficient of θ is 3.

Correct option: B**Final answer** $2\pi/3$

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Solution**

1. The outside factor controls the range.

Correct option: C**Final answer** $[-4, 4]$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Solution**

1. Rewrite $2(\theta - \pi/6)$.

Correct option: B**Final answer**right by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Solution**

1. The coefficient is $1/2$.

Correct option: D**Final answer** 4π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:

Solution

1. Factor b and read q/b first.

Correct option: B

Final answer

horizontal shift

Q060 MCQ

The period of $y = \tan(4\theta + q)$ is:

Solution

1. Tangent period is $\pi/|4|$.

Correct option: A

Final answer

$\pi/4$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Solution**

1. The absolute sine is nonnegative and the factor is 3.

Correct option: B**Final answer**

$$0 \leq y \leq 3$$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.**Solution**

1. Use the QIV reference angle and cosine's positive sign.

Final answer

$$\cos \frac{\pi}{6}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos(\frac{\pi}{2} - \theta)$.**Solution**

1. Replace the shifted functions by $-\sin \theta$ and $\sin \theta$.

Final answer

$$-\sin \theta - \sin \theta = -2 \sin \theta$$

Q064 Written

Simplify $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.**Solution**

1. Use the half-turn and complementary-angle identities.

Final answer

$$-\cos \theta + \sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan(-\frac{3\pi}{4})$ using an acute reference angle.**Solution**

1. The reference angle is $\pi/4$ and tangent is positive in QIII.

Final answer

1

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.**Solution**

1. Read the standard cosine cycle at quarter-period intervals.

Final answer $T = 2\pi, -1 \leq y \leq 1; (0, 1), (\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.**Solution**

1. Tangent is undefined at odd multiples of $\pi/2$.

Final answer

$$T = \pi; \theta = \pm \frac{\pi}{2}$$

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.**Solution**

1. Sine crosses the x-axis at integer multiples of π .

Final answer

$$0, \pi, 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Solution

1. Cosine is zero at odd quarter-turns.

Final answer

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Q070 Written

Find the period and range of $y = 4 \cos \theta$.

Solution

1. The outside factor changes the range, not the period.

Final answer

$$T = 2\pi, -4 \leq y \leq 4$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q071 **Written****Find the period and range of $y = \sin(5\theta)$.****Solution**

1. Divide the sine period by 5.

Final answer

$$T = \frac{2\pi}{5}, -1 \leq y \leq 1$$

Q072 **Written****Find the period and range of $y = \tan(\theta/4)$.****Solution**

1. Multiply the parent tangent period by 4.

Final answer

$$T = 4\pi, y \in \mathbb{R}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Solution

1. Read the inside coefficient and outside factor separately.

Final answer

horizontal compression by $1/2$, reflection in x-axis and vertical scale $1/2$; $T = \pi$

Q074 Written

Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.

Solution

1. $\theta + p$ shifts left and does not alter period.

Final answer

left by $\frac{\pi}{6}$, $T = 2\pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.**Solution**

1. θ -p shifts right by p.

Final answerright by $\frac{\pi}{2}$, $T = 2\pi$

Q076 Written

Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**Solution**

1. Set the bracket equal to zero and retain tangent's period.

Final answer $\theta = -\frac{\pi}{4}$; $T = \pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Solution

- Factor 4 from the angle: $4(\theta - \pi/4)$.

Final answer

right by $\frac{\pi}{4}$, $T = \frac{\pi}{2}$, $-2 \leq y \leq 2$

Q078 Written

For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.

Solution

- Rewrite the angle as $\frac{1}{3}(\theta + \pi/2)$.

Final answer

left by $\frac{\pi}{2}$, $T = 6\pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.**Solution**

1. Rewrite as $3 \tan 2(\theta + \pi/2)$.

Final answerleft by $\frac{\pi}{2}$, $T = \frac{\pi}{2}$

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.**Solution**

1. Rewrite the angle as $\frac{1}{2}(\theta - \pi/4)$.

Final answerright by $\frac{\pi}{4}$, $T = 4\pi$, $-2 \leq y \leq 2$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 **Concept Quiz · MCQ**

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:

Solution

1. Use $2\pi/2$.

Correct option: B

Final answer

$$\pi$$

Quiz MCQ 2 **Concept Quiz · MCQ**

$\sin(\theta + \pi)$ equals:

Solution

1. Use the half-turn identity.

Correct option: B

Final answer

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

The range of $y = 2 \cos \theta$ is:

Solution

1. Multiply the cosine range by 2.

Correct option: B

Final answer

$$-2 \leq y \leq 2$$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Solution

1. $\theta - p$ shifts right by p .

Correct option: B

Final answer

right by $\frac{\pi}{4}$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.

Solution

1. Rewrite as $3 \sin 2(\theta - \pi/2)$.

Final answer

right by $\frac{\pi}{2}$, $T = \pi$, $-3 \leq y \leq 3$

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Solution

1. Use the quarter-turn and half-turn identities.

Final answer

$\cos \theta - \sin \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

Solution

1. Shift the parent tangent graph right by $\pi/3$; shift its asymptotes with it.

Final answer

$$T = \pi; \theta = \frac{5\pi}{6} + k\pi$$

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Solution

1. Absolute sine has input period π ; the coefficient 3 gives $\pi/3$, and -2 reflects and scales.

Final answer

$$T = \frac{\pi}{3}, -2 \leq y \leq 0; \text{ reflected below the x-axis}$$

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $-\frac{1}{\sqrt{3}}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\min} = 2\pi, T_{\max} = 2\pi, T_{\min} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0)$, $(\pi, -1)$, $(\frac{3\pi}{2}, 0)$, $(2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q001 Written

Solve (a) $-2 \cos \theta = \sqrt{2}$ **and (b)** $\tan \theta + \sqrt{3} = 0$, giving the general solution for (b).

Solution

1. Rewrite as $\cos \theta = -\frac{\sqrt{2}}{2}$ and use QII/QIII.
2. For tangent, the reference angle is $\pi/3$ and the period is π .

Final answer

$$(a) \theta = \frac{3\pi}{4}, \frac{5\pi}{4} \quad (b) \theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$$

Q002 Written

Solve (a) $2 \cos \theta - 1 = 0$ **and (b)** $\tan \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

Solution

1. Use the unit circle for cosine and include both turns in the stated interval.
2. For tangent, repeat the $\pi/4$ reference angle after one period π .

Final answer

$$(a) \theta = \frac{\pi}{3}, \frac{5\pi}{3} \quad (b) \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q003 Written

Solve all given equations on $0 \leq \theta \leq 2\pi$: (a) $2 \sin \theta + 1 = 0$, (b) $2 \sin \theta = \sqrt{3}$, (c) $3 \sin \theta + 3 = 0$, (d) $\sqrt{2} \cos \theta + 1 = 0$, (e) $2 \cos \theta = \sqrt{3}$, (f) $2 \cos \theta - \sqrt{2} = 0$, (g) $\tan \theta + 1 = 0$, (h) $\sqrt{3} \tan \theta = 1$, (i) $\sqrt{3} \tan \theta = 3$, **and give general solutions for** (j) $2 \sin \theta + \sqrt{3} = 0$, (k) $\cos \theta + 1 = 0$, (l) $\sqrt{3} \tan \theta + 3 = 0$.

Solution

1. Put each equation into a single-function form, find the reference angle, then select the quadrants from the sign.
2. Use period 2π for sine/cosine and π for tangent; reject the endpoint 2π when the interval is half-open.

Final answer

(a) $\frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{3}, \frac{2\pi}{3}$ (c) $\frac{3\pi}{2}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{4}, \frac{7\pi}{4}$ (g) $\frac{3\pi}{4}, \frac{7\pi}{4}$ (h) $\frac{\pi}{6}, \frac{7\pi}{6}$ (i) $\frac{\pi}{3}, \frac{4\pi}{3}$ (j) $\frac{4\pi}{3} + 2n\pi, \frac{5\pi}{3} + 2n\pi$ (k) $\pi + 2n\pi$ (l) $\frac{2\pi}{3} + n\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q004 Written

Solve $\tan^2 \theta - (2 + \sqrt{3}) \tan \theta + 2\sqrt{3} = 0$ **and give the general solution.**

Solution

1. Factor the quadratic as $(\tan \theta - 2)(\tan \theta - \sqrt{3}) = 0$.
2. Apply the tangent period π to both roots.

Final answer

$$\tan \theta = 2 \text{ or } \tan \theta = \sqrt{3}; \quad \theta = \arctan(2) + n\pi \text{ or } \theta = \frac{\pi}{3} + n\pi$$

Q005 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $2 \cos \theta > -\sqrt{3}$, **(b)** $\tan \theta \leq \frac{1}{\sqrt{3}}$.

Solution

1. Solve the boundary equation first, then shade the correct arcs on the unit circle.
2. Tangent is undefined at odd multiples of $\pi/2$; those points are never included.

Final answer

$$(a) 0 \leq \theta < \frac{5\pi}{6} \text{ or } \frac{7\pi}{6} < \theta < 2\pi \quad (b) 0 \leq \theta \leq \frac{\pi}{6} \text{ or } \frac{\pi}{2} < \theta \leq \frac{7\pi}{6} \text{ or } \frac{3\pi}{2} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q006 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin \theta \leq -1$, (b) $\cos \theta > \frac{1}{2}$, (c) $\tan \theta > 1$.

Solution

1. Mark the equality angles and choose the arcs where the function has the required sign.
2. Open endpoints correspond to strict inequalities and to tangent asymptotes.

Final answer

$$(a) \frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6} \quad (b) 0 \leq \theta < \frac{\pi}{3} \text{ or } \frac{5\pi}{3} < \theta < 2\pi \quad (c) \frac{\pi}{4} < \theta < \frac{\pi}{2} \text{ or } \frac{5\pi}{4} < \theta < \frac{3\pi}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q007 Written

Solve all given inequalities for $0 \leq \theta < 2\pi$: (a) $\sin \theta > \frac{\sqrt{3}}{2}$, (b) $\sin \theta - \frac{\sqrt{2}}{2} > 0$, (c) $2 \sin \theta + \sqrt{3} \leq 0$, (d) $\cos \theta > 0$, (e) $\cos \theta - \frac{1}{2} \leq 0$, (f) $2 \cos \theta + \sqrt{2} \leq 0$, (g) $\tan \theta \leq -\frac{1}{\sqrt{3}}$, (h) $\tan \theta + \sqrt{3} > 0$, (i) $\tan \theta + 1 \geq 0$.

Solution

- For each item, solve the equality, draw the relevant horizontal line, and select the arcs satisfying the sign.
- Keep the distinction between open and closed endpoints and remove all tangent asymptotes.

Final answer

$$(a) \left(\frac{\pi}{3}, \frac{2\pi}{3}\right) (b) \left(\frac{\pi}{4}, \frac{3\pi}{4}\right) (c) \left[\frac{4\pi}{3}, \frac{5\pi}{3}\right] (d) \left[0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right) (e) \left[\frac{\pi}{3}, \frac{5\pi}{3}\right] (f) \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right] (g) \left(\frac{\pi}{2}, \frac{5\pi}{6}\right] \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right] (h) \left[0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \frac{3\pi}{2}\right) \cup \left(\frac{5\pi}{3}, 2\pi\right) (i) \left[0, \frac{\pi}{2}\right) \cup \left[\frac{3\pi}{4}, \frac{3\pi}{2}\right) \cup \left[\frac{7\pi}{4}, 2\pi\right)$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q008 Written

Solve $2 \sin^2 \theta - \cos \theta - 1 > 0$ for $0 \leq \theta < 2\pi$.**Solution**

1. Use $\sin^2 \theta = 1 - \cos^2 \theta$ to obtain $(1 - 2 \cos \theta)(1 + \cos \theta) > 0$.
2. Since $-1 \leq \cos \theta \leq 1$, the solution is $-1 < \cos \theta < 1/2$; exclude $\theta = \pi$ where the product is zero.

Final answer

$$\frac{\pi}{3} < \theta < \pi \text{ or } \pi < \theta < \frac{5\pi}{3}$$

Q009 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $2 \cos^2 \theta - 3 \sin \theta = 0$, **(b)** $2 \sin^2 \theta - 3 \cos \theta - 3 \leq 0$.**Solution**

1. For (a), replace $\cos^2 \theta$ and solve $2s^2 + 3s - 2 = 0$; reject $s = -2$.
2. For (b), replace $\sin^2 \theta$, solve $(2c + 1)(c + 1) \geq 0$, and intersect with $[-1, 1]$.

Final answer

$$(a) \theta = \frac{\pi}{6}, \frac{5\pi}{6} \quad (b) 0 \leq \theta \leq \frac{2\pi}{3} \text{ or } \theta = \pi \text{ or } \frac{4\pi}{3} \leq \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q010 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta - \sin \theta = 0$, (b) $2 \cos^2 \theta - \sin \theta = 2$, (c) $2 \sin^2 \theta - 4 < 5 \cos \theta$.

Solution

1. Set $s = \sin \theta$ or $c = \cos \theta$ after using $s^2 + c^2 = 1$.
2. Solve the resulting quadratic, reject impossible trig values, then read the unit-circle intervals.

Final answer

$$(a) \theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi \quad (b) \theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6} \quad (c) 0 \leq \theta < \frac{2\pi}{3} \text{ or } \frac{4\pi}{3} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q011 Written

Solve all given equations and inequalities for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta + \sin \theta - 1 = 0$, (b) $2 \sin^2 \theta + 3 \cos \theta = 0$, (c) $2 \sin^2 \theta - \cos \theta = 2$, (d) $2 \cos^2 \theta - 5 \cos \theta - 3 = 0$, (e) $2 \cos^2 \theta + \sin \theta - 1 = 0$, (f) $2 \cos^2 \theta + 3 \sin \theta = 3$, (g) $2 \cos^2 \theta + 3 \sin \theta \leq 0$, (h) $2 \sin^2 \theta + \cos \theta > 1$, (i) $2 \cos^2 \theta - 4 \leq 5 \sin \theta$.

Solution

1. Convert to one trig function, solve the quadratic, and apply $-1 \leq \sin \theta, \cos \theta \leq 1$.
2. For inequalities, use the sign chart in the trig variable and then translate the accepted interval(s) to θ .

Final answer

(a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (b) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (c) $\frac{2\pi}{3}, \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}$ (d) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (e) $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$ (g) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (h) $0 < \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ (i) $0 \leq \theta \leq \frac{7\pi}{6}$ or $\frac{11\pi}{6} \leq \theta < 2\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q012 Written

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

Solution

1. Subtract 1 and simplify the numerator to $\sin\theta(1 - \sin\theta)$.
2. The denominator is positive but undefined at $\theta = \pi/2$; hence $0 \leq \sin\theta \leq 1$ with that point removed.

Final answer

$$0 \leq \theta < \frac{\pi}{2} \text{ or } \frac{\pi}{2} < \theta \leq \pi$$

Q013 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, **(b)** $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Solution

1. Let $t = \theta + \alpha$, write the range of t , solve there, then subtract α .
2. For the inequality, shade one complete revolution in the shifted t-range and preserve strict endpoints.

Final answer

$$(a) \theta = \frac{\pi}{2}, \frac{11\pi}{6} \quad (b) 0 \leq \theta < \pi \text{ or } \frac{5\pi}{3} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q014 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

Solution

1. Find each shifted argument's exact range before selecting the unit-circle arcs.
2. For tangent, exclude asymptotes automatically; for strict cosine inequality, endpoints are open.

Final answer

$$(a) \theta = 0, \frac{2\pi}{3} \quad (b) \theta = \pi, \frac{13\pi}{12} \quad (c) \frac{\pi}{2} < \theta < \pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q015 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \leq \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \geq \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

Solution

1. Set $t = \theta + \alpha$, record its interval, solve the standard trig relation in that interval, then subtract α .
2. Check every endpoint against the original strict/non-strict sign and the domain $0 \leq \theta < 2\pi$.

Final answer

(a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, 2\pi$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{5\pi}{6}, \frac{11\pi}{6}$ (f) $\pi, 2\pi$ (g) $[0, \frac{\pi}{6}] \cup [\frac{\pi}{2}, 2\pi)$ (h) $[0, \frac{\pi}{2}]$ (i) $\frac{4\pi}{3} < \theta < 2\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q016 Written

Solve $4 \sin^2(\theta + \frac{\pi}{6}) \geq 1$ for $0 \leq \theta < 2\pi$.**Solution**

1. Let $t = \theta + \pi/6$; then $|\sin(t)| \geq 1/2$.
2. On the shifted range $[\pi/6, 13\pi/6)$, the accepted arcs give the stated closed θ -intervals.

Final answer

$$0 \leq \theta \leq \frac{2\pi}{3} \text{ or } \pi \leq \theta \leq \frac{5\pi}{3}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q017 Written

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2 \cos x}$. Evaluate this strategy and propose a better substitution on $[0^\circ, 360^\circ[$.

Solution

1. Replace $\tan(x)$ by $\sin(x)/\cos(x)$ before cancelling; the original tangent already excludes $\cos(x) = 0$.
2. Solve the resulting sine equation on the stated degree interval.

Final answer

Dividing by $\cos(x)$ separates the interacting factors and is not helpful; use $\tan(x) \cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30^\circ, 150^\circ$.

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q018 MCQ

The period used for a general tangent solution is:**Solution**

1. Tangent repeats after π .

Correct option: B**Final answer** π

Q019 MCQ

The solutions of $2 \cos \theta = 1$ on $[0, 2\pi]$ are:**Solution**

1. Cosine is $1/2$ in QI and QIV.

Correct option: B**Final answer** $\pi/3, 5\pi/3$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q020 MCQ

Which root is impossible when solving $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$?

Solution

1. A sine value must lie in $[-1, 1]$.

Correct option: B

Final answer

$$\sin \theta = -2$$

Q021 MCQ

The quadratic challenge factors into roots:

Solution

1. Use $(u - 2)(u - \sqrt{3})$ with $u = \tan \theta$.

Correct option: B

Final answer

$$2, \sqrt{3}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q022 MCQ

For $2 \cos \theta > -\sqrt{3}$, the excluded arc is:

Solution

1. Cosine is at most $-\sqrt{3}/2$ on the central QII arc.

Correct option: B

Final answer

$[5\pi/6, 7\pi/6]$

Q023 MCQ

The inequality $\tan \theta > 1$ on $[0, 2\pi)$ holds on:

Solution

1. Use the increasing tangent branches and omit asymptotes.

Correct option: A

Final answer

$(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q024 MCQ

A closed endpoint in a trig inequality is used when the sign is:**Solution**

1. Equality is included for $\leq$ or $\geq$.

Correct option: C**Final answer**

greater than or equal to

Q025 MCQ

The 5-14 challenge reduces to which cosine condition?**Solution**

1. Factor the expression after replacing $\sin^2\theta$.

Correct option: B**Final answer** $-1 < \cos \theta < 1/2$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q026 MCQ

When a quadratic gives $\sin \theta = -2$, we:**Solution**

1. Sine cannot be outside $[-1, 1]$.

Correct option: B

Final answer

reject it

Q027 MCQ

The first step for $2 \cos^2 \theta - 3 \sin \theta = 0$ is to use:**Solution**

1. Rewrite in one trig function.

Correct option: A

Final answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q028 MCQ

The solution of $2 \sin^2 \theta + \sin \theta - 1 = 0$ includes:**Solution**

1. The valid root $\sin \theta = 1/2$ gives $\pi/6, 5\pi/6$.

Correct option: A**Final answer** $\pi/6$

Q029 MCQ

In the rational challenge, which angle must be excluded?**Solution**

1. The denominator $\cos^2 \theta$ is zero there.

Correct option: C**Final answer** $\pi/2$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \leq \theta < 2\pi$, the t-range is:

Solution

1. Add $\pi/3$ to both domain endpoints.

Correct option: B

Final answer

$[\pi/3, 7\pi/3)$

Q031 MCQ

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:

Solution

1. Set the shifted argument to $\pi/3$.

Correct option: A

Final answer

$\pi/12$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ -solution is:

Solution

1. The shifted sine is at least $\sqrt{2}/2$ from $\pi/4$ to $3\pi/4$.

Correct option: A

Final answer

$$[0, \pi/2]$$

Q033 MCQ

The 5-16 challenge is equivalent to:

Solution

1. Divide by 4 and take the square-root inequality.

Correct option: A

Final answer

$$|\sin t| \geq 1/2$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q034 MCQ

In the Reflect problem, $\tan x \cos x$ simplifies to:**Solution**1. Use $\tan(x) = \sin(x)/\cos(x)$ where defined.

Correct option: B

Final answer $\sin x$

Quiz MCQ 1 Concept Quiz · MCQ

The general solution of $\tan \theta = -1$ is:**Solution**1. Tangent has period π .

Correct option: B

Final answer

$$\theta = -\frac{\pi}{4} + n\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 2 Concept Quiz · MCQ

The solution of $\cos \theta > 0$ on $[0, 2\pi)$ is:

Solution

1. Cosine is positive in QI and QIV.

Correct option: B

Final answer

$$[0, \pi/2) \cup (3\pi/2, 2\pi)$$

Quiz MCQ 3 Concept Quiz · MCQ

The valid sine root of $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ is:

Solution

1. The other algebraic root is outside the sine range.

Correct option: B

Final answer

$$\sin \theta = \frac{1}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 4 **Concept Quiz · MCQ**

The first operation in a shifted-argument problem is to:

Solution

1. The t-range prevents losing or adding solutions.

Correct option: B

Final answer

set t equal to the whole argument and find its range

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $2 \cos \theta = \sqrt{3}$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Cosine is $\sqrt{3}/2$ in QI and QIV.

Final answer

$$\theta = \frac{\pi}{6}, \frac{11\pi}{6}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\tan \theta \leq -1$ for $0 \leq \theta < 2\pi$.**Solution**

1. Use the two tangent branches and include the equality angles.

Final answer

$$\frac{\pi}{2} < \theta \leq \frac{3\pi}{4} \text{ or } \frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$$

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $2 \sin^2 \theta + \sin \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.**Solution**

1. Factor to $(2 \sin \theta - 1)(\sin \theta + 1) = 0$, then use the unit circle.

Final answer

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 8 Concept Quiz · Written

Solve $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Set $t = \theta + \pi/3$, solve $\cos(t) > 1/2$ in $[\pi/3, 7\pi/3)$, then subtract.

Final answer

$$\frac{4\pi}{3} < \theta < 2\pi$$

Answer key

Use the question number shown on each page.

- Q001** (a) $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$ (b) $\theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$ **Q012** $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$ **Q020** B
- Q002** (a) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ **Q013** (a) $\theta = \frac{\pi}{2}, \frac{11\pi}{6}$ (b) $0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$ **Q021** B
- Q003** (a) $\frac{3\pi}{4}, \frac{11\pi}{4}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q014** (a) $\theta = 0, \frac{2\pi}{3}$ (b) $\theta = \pi, \frac{13\pi}{12}$ (c) $\frac{\pi}{2} < \theta < \pi$ **Q022** B
- Q004** $\tan \theta = 2$ or $\tan \theta = \sqrt{3}$; $\theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$ **Q015** (a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, \frac{5\pi}{12}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q023** A
- Q005** (a) $0 \leq \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi$ (b) $0 \leq \theta \leq \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \leq \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$ **Q016** $0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$ **Q024** C
- Q006** (a) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (b) $0 \leq \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi$ (c) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$ **Q017** Dividing by $\cos(x)$ separates the interacting factors and is not helpful, use $\tan(x) \cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30^\circ, 150^\circ$. **Q025** B
- Q007** (a) $\frac{\pi}{2}, \frac{3\pi}{2}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q026** B
- Q008** $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$ **Q027** A
- Q009** (a) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ (b) $0 \leq \theta \leq \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \leq \theta < 2\pi$ **Q028** A
- Q010** (a) $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$ (b) $\theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $0 \leq \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ **Q029** C
- Q011** (a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (d) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (e) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (g) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (h) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (i) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (j) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (k) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (l) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (m) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (n) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (o) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (p) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (q) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (r) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (s) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (t) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (u) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (v) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (w) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (x) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (y) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (z) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ **Q018** B
- Q019** B

Q032 A

Q033 A

Q034 B

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Quiz WRITTEN 6 $\frac{\pi}{2} < \theta \leq \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$

Quiz WRITTEN 7 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Quiz WRITTEN 8 $\frac{4\pi}{3} < \theta < 2\pi$

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Worked Solutions

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Solution

1. Put $t = \cos\theta$: $y = -t^2 + t + 2 = -(t - \frac{1}{2})^2 + \frac{9}{4}$.
2. Check the vertex and the endpoints $t = -1, 1$, then return to θ .

Final answer

$$\max y = \frac{9}{4} \text{ at } \theta = \frac{\pi}{3}, \frac{5\pi}{3}; \quad \min y = 0 \text{ at } \theta = \pi$$

Q002 Written

Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Solution

1. Put $t = \sin\theta$: $y = 1 - t^2 - t = -(t + \frac{1}{2})^2 + \frac{5}{4}$.
2. The endpoint $t = 1$ gives the minimum -1 .

Final answer

$$\max y = \frac{5}{4} \text{ at } \theta = \frac{7\pi}{6}, \frac{11\pi}{6}; \quad \min y = -1 \text{ at } \theta = \frac{\pi}{2}$$

Worked Solutions

Advanced Theorems and Applications

Q003 Written

Solve the four given maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the given challenge $y = \cos^2 \theta - \sin \theta$.

Solution

1. Choose the single trig variable that makes the expression quadratic.
2. The vertex is valid only when it lies in $[-1, 1]$; convert every extremal t back to the corresponding angles.

Final answer

See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.

Worked Solutions

Advanced Theorems and Applications

Q004 Written

Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.

Solution

1. Rewrite $75^\circ = 45^\circ + 30^\circ$ and $\pi/12 = 45^\circ - 30^\circ$.
2. Apply the sine-sum and cosine-difference identities.

Final answer

$$\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ in QII and $\cos \beta = -\frac{3}{5}$ in QIII, find $\sin(\alpha + \beta)$ and $\cos(\alpha - \beta)$.

Solution

1. The quadrant signs give $\cos \alpha = -1/2$ and $\sin \beta = -4/5$.
2. Substitute in the addition identities without a calculator.

Final answer

$$\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}; \quad \cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$$

Worked Solutions

Advanced Theorems and Applications

Q006 Written

Complete the given addition-theorem exercises *exact values, quadrant – controlled α, β , and the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Solution

1. Convert radians to degrees only when it clarifies the reference angle.
2. Recover the missing sine/cosine value from the quadrant before applying the sum or difference identity.

Final answer

See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.

Q007 Written

Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.

Solution

1. Use $105^\circ = 45^\circ + 60^\circ$ and the tangent addition theorem.
2. For the lines, use $m_1 = 2, m_2 = -3$ in $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

Final answer

$$\tan 105^\circ = -2 + \sqrt{3}; \quad \theta = \frac{\pi}{4}$$

Worked Solutions

Advanced Theorems and Applications

Q008 Written

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Solution

1. Convert each line to slope-intercept form.
2. Use the acute angle convention $0 < \theta < \pi/2$, replacing θ_1 by $\pi - \theta_1$ when necessary.

Final answer

The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.

Worked Solutions

Advanced Theorems and Applications

Q009 **Written**

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Solution

1. Solve the two linear equations produced by the tangent formula for the unknown slope.
2. Use the y -intercepts and the line intersection to obtain each triangle area.

Final answer

See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.

Worked Solutions

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ **with** α **in QII, find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Solution

1. The quadrant gives $\cos \alpha = -2\sqrt{2}/3$.
2. Apply $\sin 2\alpha = 2 \sin \alpha \cos \alpha$, $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$, then divide.

Final answer

$$\sin 2\alpha = -\frac{4\sqrt{2}}{9}, \quad \cos 2\alpha = \frac{7}{9}, \quad \tan 2\alpha = -\frac{4\sqrt{2}}{7}$$

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Solution

1. The interval places α in QIII, so $\cos \alpha = -\sqrt{5}/3$.
2. Substitute into the three double-angle formulas.

Final answer

$$\sin 2\alpha = \frac{4\sqrt{5}}{9}, \quad \cos 2\alpha = \frac{1}{9}, \quad \tan 2\alpha = 4\sqrt{5}$$

Worked Solutions

Advanced Theorems and Applications

Q012 Written

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Solution

1. Determine the missing trig value from the interval before using the identities.
2. For 3θ , use $4\cos^3\theta - 3\cos\theta$; for 2θ , use the tangent double-angle identity.

Final answer

For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.

Q013 Written

Given $\cos \alpha = 3/5, 0 < \alpha < \pi$, find $\sin(\alpha/2), \cos(\alpha/2), \tan(\alpha/2)$, and evaluate $\cos(5\pi/8)$.

Solution

1. Use the half-angle formulas and choose signs from $0 < \alpha/2 < \pi/2$.
2. For $(5\pi/8)$, use $\cos(\pi - 3\pi/8)$.

Final answer

$$\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}, \quad \cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}, \quad \tan \frac{\alpha}{2} = \frac{1}{2}, \quad \cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$$

Worked Solutions

Advanced Theorems and Applications

Q014 Written

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Solution

1. Since $\alpha/2$ is in QI, both half-angle roots are positive.
2. Use $3\pi/8 = \pi/2 - \pi/8$ or the half-angle identity.

Final answer

$$\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}, \quad \cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}, \quad \tan \frac{\alpha}{2} = \sqrt{2}, \quad \sin \frac{3\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}$$

Q015 Written

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}$, $4\pi < A < 5\pi$, $\cos B = -\frac{\sqrt{5}}{3}$, $5\pi < B < 6\pi$.

Solution

1. Halve the interval first; its quadrant controls every sign.
2. Apply the half-angle formula again for $(A/4)$ after finding $(\cos(A/2))$.

Final answer

Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.

Worked Solutions

Advanced Theorems and Applications

Q016 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Solution

1. Use $\sin 2\theta = 2 \sin \theta \cos \theta$ and factor.
2. Use $\cos 2\theta = 2 \cos^2 \theta - 1$, solve the quadratic in $c = \cos \theta$, and map the interval.

Final answer

$$a \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}; \quad b \frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$$

Q017 Written

Solve the given equation $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Solution

1. Factor $\cos \theta (2 \sin \theta + 1) = 0$.
2. List the two cosine zeros and the two angles with $\sin \theta = -1/2$.

Final answer

$$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$$

Worked Solutions

Advanced Theorems and Applications

Q018 Written

Solve all given double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Solution

1. Choose the identity that leaves one trig function only.
2. For inequalities, use a sign chart and include or exclude boundary points exactly as stated.

Final answer

See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Solution

1. Plot $P(1, -\sqrt{3})$, find $r = OP$, then read the angle α .

Final answer

$$r = 2, \quad \alpha = -\frac{\pi}{3}, \quad \sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$$

Worked Solutions

Advanced Theorems and Applications

Q020 Written

Rewrite the given expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, **and** $3 \sin \theta - \cos \theta$ **in harmonic-addition form.**

Solution

1. For $a \sin \theta + b \cos \theta$, use $r = \sqrt{a^2 + b^2}$, $\cos \alpha = a/r$, $\sin \alpha = b/r$.

Final answer

$$\sqrt{2} \sin \theta - \frac{\pi}{4}, \sqrt{2} \sin \theta + \frac{\pi}{4}, 2\sqrt{3} \sin \theta - \frac{\pi}{6}, \sqrt{10} \sin \theta - \arctan \frac{1}{3}$$

Q021 Written

Rewrite all given harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, **including** $2 \sin \theta + 2 \cos \theta$ **and the challenge** $\sin \theta - \sqrt{3} \cos \theta$.

Solution

1. Keep the sign of the cosine coefficient when locating P(a,b).
2. Substitute the obtained (r,a) and expand once to verify both coefficients.

Final answer

See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.

Worked Solutions

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.**Solution**

1. Synthesize each left-hand side and account for the shifted angle range.
2. The maximum of $\sin \theta + \cos \theta$ is $\sqrt{2} < 3/2$, so the second inequality is always true.

Final answer

$$a \theta = \frac{\pi}{2}, \pi; \quad b \ 0 \leq \theta < 2\pi$$

Q023 Written

Solve $\sin \theta \leq \cos \theta$ on $0 \leq \theta < 2\pi$.**Solution**

1. Rewrite as $\sqrt{2} \sin(\theta - \pi/4) \leq 0$.
2. Read the accepted half-turns and retain the equality endpoints.

Final answer

$$0 \leq \theta \leq \frac{\pi}{4} \quad \text{or} \quad \frac{5\pi}{4} \leq \theta < 2\pi$$

Worked Solutions

Advanced Theorems and Applications

Q024 **Written**

Solve all given harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Solution

1. After synthesis set $t = \theta + \alpha$ and write the exact t-range.
2. Use one full revolution in t, then subtract α and intersect with the original domain.

Final answer

See the numbered solution; synthesize first, then solve in the shifted variable over its full range.

Q025 **Written**

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Solution

1. Use $-1 \leq \sin(t) \leq 1$ for the first function.
2. For the second, set $t = \cos \theta$ and complete the square after the double-angle substitution.

Final answer

First: max = 2, min = -2. Second: max = 7 at $\theta = 0$; min = -2 at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Worked Solutions

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the given functions $y = -\frac{4}{3}\sin^2\theta + \cos\theta$ and $y = \cos 2\theta - 2\cos\theta$.

Solution

1. Reduce double angles or $\sin^2\theta$ to one variable.
2. Check the vertex and both endpoints of the allowed interval.

Final answer

For the first, $\max = 1$, $\min = -\frac{73}{48}$; for the second, use $t = \cos\theta$ and the endpoint/vertex test.

Q027 Written

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Solution

1. Use the same transformation rules as for equations and inequalities.
2. Never report a vertex outside $[-1, 1]$; compare it with both endpoints.

Final answer

See the numbered solution: synthesize $a\sin\theta + b\cos\theta$ or substitute $t = \cos\theta$, then test the full allowed range.

Worked Solutions

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:**Solution**1. Use $\sin^2 \theta = 1 - t^2$.

Correct option: B

Final answer $t = \cos \theta$

Q029 MCQ

The maximum of $\sin^2 \theta + \cos \theta + 1$ is:**Solution**

1. Complete the square.

Correct option: C

Final answer $9/4$

Worked Solutions

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Solution**

1. Trig sine and cosine values are bounded.

Correct option: B**Final answer** $[-1, 1]$

Q031 MCQ

The identity for $\sin(a+b)$ is:**Solution**

1. Use the addition theorem.

Correct option: A**Final answer** $\sin a \cos b + \cos a \sin b$

Worked Solutions

Advanced Theorems and Applications

Q032 MCQ

sin 75 degrees equals:**Solution**

1. Rewrite 75 as 45+30.

Correct option: B

Final answer

$$\sqrt{6} + \sqrt{2}/4$$

Q033 MCQ

A missing trig value is selected using:**Solution**

1. The quadrant fixes the sign.

Correct option: A

Final answer

the quadrant

Worked Solutions

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Solution**

1. The coefficient of x is the slope.

Correct option: B**Final answer** m

Q035 MCQ

The angle formula uses denominator:**Solution**

1. Use the tangent difference theorem.

Correct option: A**Final answer** $1 + m_1 m_2$

Worked Solutions

Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Solution

1. Choose the acute representative.

Correct option: B

Final answer

$\pi - \theta_1$

Q037 MCQ

$\sin(2a)$ equals:

Solution

1. Double-angle identity.

Correct option: B

Final answer

$2 \sin a \cos a$

Worked Solutions

Advanced Theorems and Applications

Q038 MCQ

 $\cos(2a)$ can be written as:**Solution**

1. Use the standard formula.

Correct option: A**Final answer**

$$\cos^2 a - \sin^2 a$$

Q039 MCQ

For theta in QIV, sin theta is:**Solution**

1. Read the quadrant.

Correct option: B**Final answer**

negative

Worked Solutions

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Solution**1. Choose the sign from $a/2$.

Correct option: A

Final answer $\sqrt{(1 - \cos a)/2}$

Q041 MCQ

The sign of a half-angle root is chosen from:**Solution**

1. Halving changes the interval.

Correct option: A

Final answer

the original quadrant interval

Worked Solutions

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Solution

1. $a/2$ is in QI.

Correct option: B

Final answer

positive

Q043 MCQ

For double-angle equations, first rewrite in:

Solution

1. Use a double-angle identity.

Correct option: A

Final answer

one trig function

Worked Solutions

Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Solution**1. Factor after substituting $\sin(2\theta)$.

Correct option: A

Final answer $\sin\theta(2\cos\theta + 1)$

Q045 MCQ

A tangent asymptote is always:**Solution**

1. Tangent is undefined there.

Correct option: B

Final answer

excluded

Worked Solutions

Advanced Theorems and Applications

Q046 MCQ

For a $\sin \theta + b \cos \theta$, r is:**Solution**

1. It is the length OP.

Correct option: B**Final answer**

$$\sqrt{a^2 + b^2}$$

Q047 MCQ

The harmonic target form is:**Solution**

1. Match the coefficients.

Correct option: A**Final answer**

$$r \sin(\theta + a)$$

Worked Solutions

Advanced Theorems and Applications

Q048 MCQ

The angle a is located from point:**Solution**

1. The coefficient point determines direction.

Correct option: A

Final answer $P(a, b)$

Q049 MCQ

After synthesis, t is usually:**Solution**

1. Track the shifted range.

Correct option: A

Final answer $\theta + a$

Worked Solutions

Advanced Theorems and Applications

Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < 3/2$ is:

Solution

1. $\sqrt{2} < 3/2$.

Correct option: B

Final answer

always true

Q051 MCQ

In a strict inequality, equality endpoints are:

Solution

1. Use open endpoints.

Correct option: B

Final answer

excluded

Worked Solutions

Advanced Theorems and Applications

Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Solution**

1. Multiply the sine range by 2.

Correct option: B

Final answer $[-2, 2]$

Q053 MCQ

For a double-angle maximum problem, use:**Solution**

1. Reduce to $t = \sin \theta$ or $\cos \theta$.

Correct option: A

Final answer

the double-angle formula then a bounded variable

Worked Solutions

Advanced Theorems and Applications

Q054 MCQ

A vertex outside $[-1,1]$ is:**Solution**

1. It is not attainable by sine/cosine.

Correct option: B**Final answer**

rejected

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:**Solution**

1. Use the sine range.

Correct option: D**Final answer**

2

Worked Solutions

Advanced Theorems and Applications

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The addition-theorem value $\sin 75^\circ$ is:

Solution

1. Use 45 degrees plus 30 degrees.

Correct option: A

Final answer

$$\sqrt{6} + \sqrt{2}/4$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Solution

1. Complete the square.

Correct option: A

Final answer

$$-2$$

Worked Solutions

Advanced Theorems and Applications

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Solution

1. The tangent magnitude is 1.

Correct option: B

Final answer

$$\pi/4$$

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

Solution

1. Let $t = \cos \theta$ and complete the square.

Final answer

$$\max = \frac{9}{4}, \min = 0$$

Worked Solutions

Advanced Theorems and Applications

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ **when** $\sin \alpha = 1/3$ **and** α **is in** QII.

Solution

1. Use the quadrant to select $\cos \alpha$.

Final answer

$$-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$$

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

Solution

1. Find r and the coefficient angle from $P(3, -\sqrt{3})$.

Final answer

$$2\sqrt{3} \sin \theta - \frac{\pi}{6}$$

Worked Solutions

Advanced Theorems and Applications

Quiz WRITTEN 8 **Concept Quiz · Written****Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.**Solution**1. Synthesize to $\sqrt{2} \sin(\theta - \pi/4) \leq 0$.**Final answer**

$$\left[0, \frac{\pi}{4}\right] \cup \left[\frac{5\pi}{4}, 2\pi\right)$$

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \pi, 2\pi$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge,
 $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$,
 $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b \frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{2}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a \theta = \frac{\pi}{2}, \pi$; $b \ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q025** First: $\max = 2$, $\min = -2$. Second: $\max = 7$ at $\theta = 0$; $\min = -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q026** For the first, $\max = 1$, $\min = -\frac{73}{48}$; for the second, use $t = \cos \theta$ and the endpoint/vertex test.
- Q027** See the numbered solution: synthesize $a \sin \theta + b \cos \theta$ or substitute $t = \cos \theta$, then test the full allowed range.
- Q028** B
- Q029** C
- Q030** B
- Q031** A
- Q032** B
- Q033** A
- Q034** B
- Q035** A
- Q036** B
- Q037** B
- Q038** A
- Q039** B
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}$, $\min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$